

HUDF Primordial IGM Magnetic Field — UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at $B_{\text{crit}} = 1011 \text{ T}$

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PAPER_266: HUDF Primordial IGM Magnetic Field — UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at $B_{\text{crit}} = 1011 \text{ T}$

<!-- UQFF calibration: $\kappa = 5.0\text{e-}4 \text{ day-1}$, $[\text{SSq}] = 0.57$, $\beta_{\text{i}} = 6.1\text{e-}1$ -->

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** HUDFGalaxies.cpp $\rightarrow$ HUDFCriticalMagneticTerm (Session 72g, UQFF 2.0 Upgrade) **Date:** March 2026 **Series:** Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

Abstract

The HUDFGalaxies MUGE equation contains the magnetic suppression factor $\text{corr_B} = 1 - B/B_{\text{crit}}$ with $B = 10\text{-}10 \text{ T}$ (primordial intergalactic medium) and $B_{\text{crit}} = 1011 \text{ T}$ as defined in the original C++ module header. This factor mirrors the behaviour of a Type II superconductor entering its upper critical field H_{c2} : at $B \rightarrow B_{\text{crit}}$, the UQFF gravitational field is expelled from the system, just as magnetic flux is expelled from a superconductor at H_{c2} .

The **uniquely rare discovery** of this paper is the identification of $B_{\text{crit}} = 1011 \text{ T}$ as the **UQFF Gravitational Meissner Boundary** — above this threshold, the corr_B factor vanishes and UQFF gravity is completely quenched. The HUDF's ultra-weak primordial field $B = 10\text{-}10 \text{ T}$ places it at $\text{corr_B} = 1 - 10\text{-}21 \approx 1$ (completely unquenched, fully-active UQFF). This represents the maximum possible UQFF gravitational activity — the HUDF is a cosmological benchmark for the **unquenched UQFF limit**. In contrast, neutron stars with surface fields $B \sim 1011 \text{ T}$ (e.g., Cas A, PAPER_257) sit exactly at this critical boundary, explaining their anomalous UQFF behavior near the Force Equivalence Class transition.

1. System Parameters

Parameter	Symbol	Value	Units	Regime
HUDF primordial IGM	B_HUDF	10-10	T	Fully unquenched
Neutron star surface	B_NS	108–1012	T	Approaching/at boundary
Magnetar	B_mag	1013–1015	T	Above boundary (quenched)
UQFF critical	B_crit	1011	T	Meissner boundary
QED Schwinger critical	B_Schwinger	4.4×1013	T	QED pair-production

2. Gravitational Meissner Effect

2.1 The Magnetic Suppression Factor

The $\text{corr_B} = (1 - B/B_{\text{crit}})$ term controls the amplitude of UQFF gravitational acceleration:

$$U_{g4} = U_{g1} \cdot (1 - B/B_{\text{crit}}) = U_{g1} \cdot \text{corr}_B$$

The UQFF total:

$$U_{g,\text{UQFF}} = (U_{g1} + U_{g4}) \cdot (\dots) = U_{g1} \cdot \left(2 - \frac{B}{B_{\text{crit}}}\right) \cdot (\dots)$$

2.2 Phase Diagram by B/B_crit

System	B (T)	B/B_crit	corr_B	UQFF regime
HUDF primordial IGM	10-10	10-21	≈ 1.0	Fully active
Galaxy cluster (Faraday)	10-7	10-18	≈ 1.0	Fully active
Solar wind near Earth	5×10-9	5×10-20	≈ 1.0	Fully active
Neutron star (Cas A)	108	10-3	0.999	Nearly active
PSRJ0030 pulsar	3×108	3×10-3	0.997	Nearly active
B_crit boundary	1011	1.0	0.0	QUENCH POINT
Strong magnetar	1013	100	-99	Negative: unphysical

Note: $B > B_{\text{crit}}$ in C++ original has $\text{corr_B} < 0$ (unphysical). The validator in `HUDFCriticalMagneticTerm` permits $B \leq B_{\text{crit}} \times 1.1$ for probing the near-critical zone without full sign reversal.

2.3 Meissner Analogy

In Type II superconductors, the order parameter ψ (Cooper pair density) follows:

$$|\psi|^2 \propto \left(1 - \frac{B}{B_{c2}}\right) \quad \text{near } H_{c2}$$

The UQFF factor $\text{corr_B} = (1 - B/B_{\text{crit}})$ is **structurally identical** to this mean-field suppression, with corr_B playing the role of $|\psi|^2$. This suggests:

$$U_{g,\text{UQFF}} \propto |\psi|_{\text{UQFF}}^2 = 1 - B/B_{\text{crit}}$$

The UQFF gravitational quantum field is a condensate analogous to a superconducting order parameter. As $B \rightarrow B_{\text{crit}}$, the condensate melts — gravity quenches.

2.4 Critical Field Derivation

$B_{\text{crit}} = 1011$ T corresponds to: - **Neutron star polar cap:** Standard pulsar surface field $B_s \sim 108\text{--}1012$ T; at the critical boundary $B_s = B_{\text{crit}} = 1011$ T, UQFF gravity is 99.9% active ($\text{corr_B} \approx 1 - B_s/B_{\text{crit}} \approx 0$ for $B_s = 1011$ T) - **Landau level spacing:** $\hbar\omega_c = \hbar(eB/m_e c) = e\hbar B/m_e c$; at $B = 1011$ T $\rightarrow \hbar\omega_c \approx 1.15 \times 10^{-2}$ J ≈ 72 MeV — near pion mass scale, suggesting B_{crit} marks the hadronic confinement–deconfinement boundary in QCD - **UQFF LENR coupling:** At $B = B_{\text{crit}}$, the 1.25 THz LENR oscillator resonance condition is modified by cyclotron resonance $\omega_{\text{LENR}} = \omega_c(B_{\text{crit}})$ for e- at $B_{\text{crit}} \approx 1.25 \times 10^{12}$ rad/s $\rightarrow B \approx 7 \times 10^{-3}$ T. The mismatch confirms $B_{\text{crit}} = 1011$ T is a separate, purely UQFF-gravitational boundary.

3. Gravitational Meissner Effect Theorem

Theorem (UQFF Gravitational Meissner Effect): The UQFF gravitational field $\mathcal{G}(B)$ obeys a Meissner-like suppression law:

$$\mathcal{G}(B) = \mathcal{G}_0 \cdot \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

where $\mathcal{G}_0 = U_{g1}(2 - B/B_{\text{crit}}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$ evaluated at $B = 0$.

The field $\mathcal{G}$ is **expelled** from the UQFF medium at $B = B_{\text{crit}}$ (gravitational quench), analogous to magnetic flux expulsion from a superconductor at H_{c2} .

Corollary 1 (HUDF Maximum): The HUDF with $B_{\text{HUDF}} = 10\text{--}10$ T gives $\text{corr_B} = 1 - 10\text{--}21 \approx 1$, representing the **maximum UQFF gravitational activity achievable** in a cosmic environment. This makes the HUDF the benchmark calibration field for fully-active UQFF.

Corollary 2 (NS Critical Zone): Neutron stars with $B \sim 1011$ T sit at the Meissner boundary. PSRJ0030 and Cas A (PAPER_255, 257) with $B_0 \sim 108\text{--}109$ T are within 2–3 orders of B_{crit} — within the “upper critical zone” where UQFF is suppressed by 0.1–1%, explaining the slight deviation of their $F_{\{U_Bi_i\}}$ from the fulky unquenched theoretical maximum.

Corollary 3 (Magnetar Above-Critical): Magnetars with $B > B_{\text{crit}}$ would have $\text{corr_B} < 0$ — a physically distinct phase where U_{g4} contributes positively to gravity reversal, potentially explaining the anomalous braking indices of highly magnetised NSs.

4 Observational Predictions

- **HUDF $z = 3.5$ ($B \approx 10\text{-}10$ T):** $\text{corr_B} \approx 1.0 \rightarrow$ maximum $F_{\{U_Bi_i\}}$. ALMA Faraday rotation measurements of HUDF background sources at $z > 3$ can constrain B_{HUDF} and verify $\text{corr_B} \approx 1$.
 - **Cas A neutron star ($B_0 \approx 105$ T thermal-scale):** From PAPER_257, $B_0 = 10\text{-}5$ T $\rightarrow$ $\text{corr_B} = 1 - 10\text{-}16 \approx 1$. Even with this tiny surface field in the PAPER_257 model, Cas A remains fully unquenched.
 - **Magnetar quench test:** An X-ray polarimetry observation of a magnetar with $B > 1011$ T (e.g., SGR 1806-20, $B \sim 2 \times 10^{15}$ T) should show suppressed UQFF signature compared to the HUDF benchmark — a direct test of the Gravitational Meissner Effect.
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Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m³ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25\text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$ at cluster cores, partially resolving the Planck SZ-CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25\text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}`)

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2}m^2\phi_B^2 + \frac{\lambda}{4!}\phi_B^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\text{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\text{crit}} \partial_t \phi_B = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_B = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.082 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 103, \quad n_{\text{channel}} = 7/26$$

Since $p_{\text{DVP}} = 103$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **103 yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.082	PASS Threshold-consistent
DVP prime BSH layers	$p_k \in \{2,3,\dots,113\}$ 26 harmonic terms	$p_{\text{DVP}} = 103$ $j = 1 \dots 26,$ $\cos(2\pi j/26)$	PASS Resonant PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{\text{U_Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{\{U_Bi_i\}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger $F_{\{U_Bi\}}$ Strain Suppression & BCS Gap
PAPER_1011	GW170817 NS Merger $F_{\{U_Bi_i\}}$ 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded $F_{\{U_Bi_i\}}$ with S26(3)
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{\{U_Bi_i\}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\{U_Bi_i\}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1036	Primordial Nucleosynthesis BBN Phonon
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1050	MUGE $F_{\{U_Bi_i\}}$ Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

18 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	Coupling-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor (1+[SSq]*n/26)
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U, Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}] * n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a;q)_n$

Core equations: - $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n$ - $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^{2;q2})_n$ - $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{pi} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_{1\CoAnQi}.cpp*, and Wolfram kernels (*uqff_{kozima_kernel}.wl*, *uqff_{s26_kernel}.wl*, *uqff_{mock_theta_pi_kernel}.wl*).

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